

Problem 26.4

Find the present value of a perpetuity-immediate whose successive payments are \$5, \$10, \$15, $\dots$ assuming an annual effective rate of interest of 5%.

Solution

The payments are

$$5, 10, 15, \dots$$

which form an increasing perpetuity-immediate. This can be written as

$$5(Ia)_{\infty|}.$$

For an increasing perpetuity-immediate,

$$(Ia)_{\infty|} = \frac{1}{i} + \frac{1}{i^2}.$$

Since

$$i = 0.05,$$

the present value is

$$PV = 5 \left(\frac{1}{0.05} + \frac{1}{0.05^2} \right).$$

Thus,

$$PV = 5(20 + 400) = 2100.$$

Therefore, the present value is

$$\boxed{2100}.$$